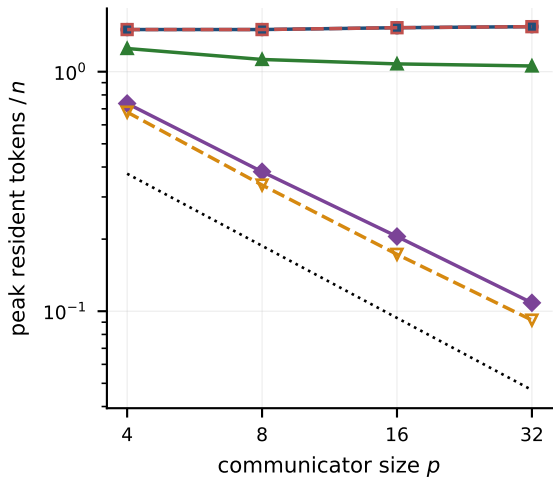


(a) residency vs. communicator size



allreduce, recursive doubling

allreduce, binomial tree

allreduce, linear

reduce-scatter, ring

reduce-scatter, recursive halving

$\propto 1/p$

(b) absolute residency, 256 entries/rank

